

Debug With Shubham Notes

TCS NQT Test Curriculum 2025

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TCS NQT Foundation Aptitude Curriculum

Number System

Mensuration

Arithmetic Ability

Elementary Statistics

Data Interpretation

Simplification & Approximation

Number System

1- Divisibility

2- Number System , HCF & LCF

3- Numbers and Decimal Frations

Formulas of Number System:

1. $1 + 2 + 3 + 4 + 5 + \dots + n = \frac{n(n + 1)}{2}$
2. $(1^2 + 2^2 + 3^2 + \dots + n^2) = \frac{n(n + 1)(2n + 1)}{6}$
3. $(1^3 + 2^3 + 3^3 + \dots + n^3) = \left(\frac{n(n + 1)}{2}\right)^2$
4. Entirety of first n odd numbers $= n^2$

Mathematical Formulas to solve questions

1. $(a + b)(a - b) = (a^2 - b^2)$
2. $(a + b)^2 = (a^2 + b^2 + 2ab)$
3. $(a - b)^2 = (a^2 + b^2 - 2ab)$
4. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
5. $(a^3 + b^3) = (a + b)(a^2 - ab + b^2)$
6. $(a^3 - b^3) = (a - b)(a^2 + ab + b^2)$
7. $(a^3 + b^3 + c^3 - 3abc) = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$
8. when $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$

Types of Number System:

Natural Numbers	Complex Number
Whole Numbers	Imaginary Number
Integers	Even Number
Real Number	Odd Number
Rational Number	Prime Number
Irrational Number	Composite Number

Whole Number 0, 1, 2, 3, 4, 5...

Natural Number 1, 2, 3, 4, 5, 6...

Integers ...-3, -2, -1, 0, 1, 2, 3, 4, 5,...

Prime Number 3, 5, 7, 11, 13, 17.....

Co-Prime Number HCF = 1

Composite Number 4, 6, 8, 9, 12, 14, 15.....

Even Number 2, 4, 6, 8, 10...

Odd Number 1, 3, 5, 7, 9....

Rational Number: 1.23, 0.3333

Irrational Number : $\sqrt{2}$, π (pi), e (Euler's number)

Complex Number : $2+3i$

Imaginary Number: $i^2 = -1$, $i^3 = -i$

Formulas for finding the Squares of a number.

Squares of numbers between 91-100:

$$97^2$$

$$91^2$$

Squares of numbers between 100-109:

$$102^2$$

$$107^2$$

Squares of numbers between: 51-60:

$$53^2$$

$$42^2$$

Important Formulas to solve Number System quickly

$$a^m * a^n =$$

1. $(a + b)(a - b) = (a^2 - b^2)$

2. $(a + b)^2 = (a^2 + b^2 + 2ab)$

3. $(a - b)^2 = (a^2 + b^2 - 2ab)$

4. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

5. $(a^3 + b^3) = (a + b)(a^2 - ab + b^2)$

6. $(a^3 - b^3) = (a - b)(a^2 + ab + b^2)$

7. $(a^3 + b^3 + c^3 - 3abc) = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$

8. when $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$

Question 1: If x and y are odd numbers, then which of the following is always an even number?

A. $x+y$

B. xy

C. x/y

D. $xy+2$

Question 2: How many prime numbers are there from 1 to 50?

A. 20

B. 15

C. 21

D. 25

Ans- 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47

Question 3: $1198 * 1198 = ?$

A. 1435204

B. 1432504

C. 1453204

D. 1435024

Solution: $(1198)^2$

$$(1200 - 2)^2$$

We can use the identity: $(a-b)^2 = a^2 + b^2 - 2ab$

$$(1200)^2 + 2^2 - 2 * 1200 * 2$$

$$1440000 + 4 - 4800$$

$$1435204$$

Question 4: The sum of first five prime numbers is:

A. 20

B. 28

C. 30

D. 25

Solution : Sum of first five prime numbers is =

$$2+3+5+7+11 = 28$$

Question 1. Find the L.C.M of 15, 30, 45?

- A. 90**
- B. 95
- C. 92
- D. None of the above

Question 2: The given ratio of two numbers is 3:2. If the L.C.M of them is 30, then calculate their sum.

- A. 34
- B. 25**
- C. 55
- D. None of the above

Question3: Find the L.C.M of 25, 35, and 55

- A. 1900
- B. 1990
- C. 1925**
- D. None of the above

Question 4. Calculate the HCF of 22 and 33

- A. 11**
- B. 12
- C. 15
- D. None of the above

Question5: If 20 is the HCF of two particular numbers and the other two factors of their LCM are 10 and 12, find the larger number?

- A. 220
- B. 210
- C. 240**
- D. None of the above

Question 6. The HCF of three specific numbers 6, 12, and 18 is 24 , find the LCM?

- A. 55
- B. 54**
- C. 67
- D. None of the above

Question 7: The two specific numbers are in the ratio 6:7, if the HCF of the given numbers is 30, what will be the numbers?

A. 160, 180

B. 180, 210

C. 200, 160

D. None of the above

Question 8: Determine the largest length of the tape, which can measure tape of 5 cm, 7cm, and 13 cm?

A. 1

B. 7

C. 13

D. None of the above

Question 9 : The HCF of two numbers is 45, and their LCM is 90, if one specific number is 9, find the other number.

A. 450

B. 435

C. 426

D. None of the above

Questions 10: Street lamps start at the interval of 5, 10, 15, 20, and 25 seconds, calculate the number in which the street lamps began together in 40 minutes

- A. 9 times
- B. 10 times
- C. 8 times
- D. None of the above

Question 11: Which of the following numbers must be added to 5678 to give a remainder of 35 when divided by 460?

- A. 980
- B. 955
- C. 797
- D. 618

Questions 12: A number when successively divided by 5, 2, 3 gives remainder 0, 1, 2 respectively in that order. What will be the remainder when the same number is divided successively by 2, 3, 5 in that order?

- A. 4,3, 2
- B. 1,0,4
- C. 2,1,3
- D. 4,1,2

Question 13: Find the least number which when divided by 12, 27 and 35 leaves 6 as a remainder?

- A. 3586
- B. 3756
- C. 3786
- D. 4786

Questions 14: What will be the reminder when $(1234567890123456789)_{24}$ is divided by 6561?

- A. 7
- B. 5
- C. 0
- D. 1

Questions 15: The remainder when $m + n$ is divided by 12 is 8, and the remainder when $m - n$ is divided by 12 is 6. If $m > n$, then what is the remainder when mn divided by 6?

- A. 3
- B. 4
- C. 2
- D. 1

Questions 16: A number 2088 by what if we divide the number it will become a perfect square?

A. 58

B. 20

C. 28

D. 29

Question 17: For any two numbers we define an operation \$ yielding another number, $X \$ Y$ such that following condition hold. s:

• $X \$ X = 0$ for all X

Find the Value of $2012 \$ 0 + 2012 \$ 1912$?

A. 2112

B. 100

C. 5936

D. Can not be determined

Question 18: If $A = x^3y^2$ and $B = xy^3$ then find the HCF of A, B?

A. $x^4 y^4$

B. xy^2

C. xy

D. X^3

Questions 19: In how many way possible ways can you write 1800 as a product of 3 positive integers X, Y, Z?

A. 350

B. 360

C. 380

D. 450

Questions 20: Find the greatest number that will divide 148 246 and 623 leaving remainders 4 6 and 11 respectively?

A. 20

B. 12

C. 6

D. 48

Solution for LCM and HCM

Ans: 1

$$2 \mid 15, 30, 45$$

$$3 \mid 15, 15, 45$$

$$3 \mid 5, 5, 15$$

$$5 \mid 5, 5, 5$$

$$1 \mid 1, 1, 1$$

$$\text{L.C.M} = 2 \times 3 \times 3 \times 5 = 90$$

Ans : 3

$$5 \mid 25, 35, 55$$

$$5 \mid 5, 7, 11$$

$$7 \mid 1, 7, 11$$

$$11 \mid 1, 1, 11$$

$$1 \mid 1, 1, 1$$

$$\text{L.C.M} = 5 \times 5 \times 7 \times 11 = 1925$$

Ans- 2

Step 1: Let's represent the two numbers as $3x$ and $2x$, where x is some integer.

Step 2: We know that for any two numbers a and b :
 $\text{LCM}(a, b) \times \text{GCD}(a, b) = a \times b$

Step 3: In this case, $a = 3x$ and $b = 2x$
 $30 \times \text{GCD}(3x, 2x) = 3x \times 2x$

Step 4: Simplify the right side of the equation
 $30 \times \text{GCD}(3x, 2x) = 6x^2$

Step 5: The GCD of $3x$ and $2x$ is x (since 3 and 2 are coprime)
 $30 \times x = 6x^2$

Step 6: Solve for x
 $30 = 6x$
 $x = 5$

Step 7: Now we can find our two numbers
First number $= 3x = 3 \times 5 = 15$
Second number $= 2x = 2 \times 5 = 10$

Step 8: Calculate the sum of the two numbers
Sum $= 15 + 10 = 25$

Therefore, the sum of the two numbers is 25.

Ans : 4

$$22 = 2 \times 11$$

$$33 = 3 \times 11$$

So the HCF will be 11

Ans : 6

$$(6 \times 12 \times 18) / 24$$

The next number will be = 54

ans: 7

Let the numbers be $6y$ and $7y$

$$\text{HCF} = 30$$

$$\text{The numbers will be } 6 \times 30 = 180$$

$$\text{And } 7 \times 30 = 210$$

Ans : 5

Step 1: Let's call the two numbers we're looking for a and b .

Step 2: We know that for any two numbers a and b :
 $\text{LCM}(a, b) \times \text{HCF}(a, b) = a \times b$

Step 3: We're told that the other two factors of the LCM are 10 and 12. This means:
 $\text{LCM}(a, b) = 20 \times 10 \times 12 = 2400$

Step 4: Now we can use the formula from Step 2:
 $2400 \times 20 = a \times b$

Step 5: Simplify:
 $48000 = a \times b$

Step 6: We know that a and b can be expressed as:
 $a = 20 \times m$
 $b = 20 \times n$
Where m and n are coprime (their HCF is 1)

Step 7: Substituting into the equation from Step 5:
 $48000 = (20m) \times (20n)$
 $48000 = 400mn$
 $120 = mn$

Step 8: The factors of 120 that are coprime are 10 and 12 (which matches our original information)

Step 9: Therefore:
 $a = 20 \times 12 = 240$
 $b = 20 \times 10 = 200$

Step 10: The larger number is 240.

Therefore, the larger number is 240.

Ans : 8

We need to find the GCD of 5 cm, 7 cm, and 13 cm.

We can use the Euclidean algorithm to find the GCD. However, since these numbers are relatively small, we can also list out the factors:

Factors of 5: 1, 5

Factors of 7: 1, 7

Factors of 13: 1, 13

The common factor among all these numbers is 1.

Ans : 9

HCF * LCM = Products of Numbers

$$45 * 90 = 9 * x$$

Another number will be = $(45*90)/9 = 450$

Ans : 10

1 First, we need to find the Least Common Multiple (LCM) of the intervals: 5, 10, 15, 20, and 25 seconds.

To find the LCM:

- First, let's prime factorize these numbers:

$$5 = 5$$

$$10 = 2 \times 5$$

$$15 = 3 \times 5$$

$$20 = 2^2 \times 5$$

$$25 = 5^2$$

- The LCM will include the highest power of each prime factor:

$$\text{LCM} = 2^2 \times 3 \times 5^2 = 300 \text{ seconds}$$

This means the street lamps will all start together every 300 seconds (or 5 minutes).

Now, we need to find out how many times this will happen in 40 minutes.

Convert 40 minutes to seconds:

$$40 \text{ minutes} = 40 \times 60 = 2400 \text{ seconds}$$

Divide the total time by the interval:

$$2400 \div 300 = 8$$

Therefore, in 40 minutes, the street lamps will start together 8 times.

Ans : 11

980:

$$(5678 + 980) \div 460 = 14 \text{ remainder } 398$$

This is not correct as the remainder is not 35.

955:

$$(5678 + 955) \div 460 = 14 \text{ remainder } 373$$

This is not correct as the remainder is not 35.

797:

$$(5678 + 797) \div 460 = 14 \text{ remainder } 215$$

This is not correct as the remainder is not 35.

618:

$$(5678 + 618) \div 460 = 13 \text{ remainder } 356$$

This is not correct as the remainder is not 35.

Interestingly, none of the given options produce a remainder of 35 when added to 5678 and divided by 460.

However, we can notice that 797 is the closest to our calculated answer of 157. In fact:

$$797 = 157 + 640$$

640 is an exact multiple of 460 ($640 = 460 * 1 + 180$)

This means that adding 797 will give the same remainder as adding 157 when divided by 460.

Ans 12:

Let us say that a number N , when successively divided by a , b , and c leaves a remainder of p , q , and r
 $\Rightarrow$ Before the last division by c , the number must have been of the format of $ck + r$. Here k is a natural number.

Same logic can be extended to give the value of N .

$$\Rightarrow N = a[b(ck + r) + q] + p$$

In this case:

$$N = 5[2(3k + 2) + 1] + 0$$

$$\Rightarrow N = 5[6k + 4 + 1]$$

$$\Rightarrow N = 5[6k + 5]$$

$$\Rightarrow N = 30k + 25$$

Now, we need to calculate the remainders when the number is successively divided by 2, 3 and 5.
The question will be simpler if I just assume some value of ' k '

Let us put $k = 1$

$$\Rightarrow N = 55$$

$$\Rightarrow 55/2 = 27 + \text{remainder of } 1$$

$$\Rightarrow 27/3 = 9 + \text{remainder of } 0$$

$$\Rightarrow 9/5 = 1 + \text{remainder of } 4$$

So, the remainders are 1, 0, and 4

Ans 13:

Let's call our number n .

According to the question:

When n is divided by 12, the remainder is 6

When n is divided by 27, the remainder is 6

When n is divided by 35, the remainder is 6

This means n can be expressed as:

$n = 12q_1 + 6$, where q_1 is some integer

$n = 27q_2 + 6$, where q_2 is some integer

$n = 35q_3 + 6$, where q_3 is some integer

For this to be true, $n - 6$ must be divisible by 12, 27, and 35.

In other words, $n - 6$ must be the least common multiple (LCM) of 12, 27, and 35.

To find the LCM, let's first find the prime factorization of these numbers:

$$12 = 2^2 \times 3$$

$$27 = 3^3$$

$$35 = 5 \times 7$$

The LCM will include the highest power of each prime factor:

$$\text{LCM} = 2^2 \times 3^3 \times 5 \times 7 = 4 \times 27 \times 35 = 3780$$

So, $n - 6 = 3780$

$$n = 3786$$

Therefore, the least number which when divided by 12, 27, and 35 leaves 6 as a remainder is 3786.

Ans : 14

First, let's consider the properties of modular arithmetic:

$$(a * b) \bmod n = ((a \bmod n) * (b \bmod n)) \bmod n$$

This means we can first find $(1234567890123456789 \bmod 6561)$, then raise this result to the 24th power, and finally take the mod 6561 again.

To find $1234567890123456789 \bmod 6561$:

We can use the fact that $6561 = 3^8$

$$1234567890123456789 \bmod 3 = 0$$

This means 1234567890123456789 is divisible by 3, so it's also divisible by $6561 (3^8)$.

Therefore, $1234567890123456789 \bmod 6561 = 0$

Now, $0^{24} = 0$

And finally, $0 \bmod 6561 = 0$

Ans 15:

Let's express m and n in terms of some quotients and remainders:

$$m = 12a + r$$

$$n = 12b + s$$

Where a and b are some integers, and r and s are the remainders when m and n are divided by 12.

Given that the remainder when $m + n$ is divided by 12 is 8:

$$(m + n) \equiv 8 \pmod{12}$$

$$(12a + r) + (12b + s) \equiv 8 \pmod{12}$$

$$r + s \equiv 8 \pmod{12}$$

Also, the remainder when $m - n$ is divided by 12 is 6:

$$(m - n) \equiv 6 \pmod{12}$$

$$(12a + r) - (12b + s) \equiv 6 \pmod{12}$$

$$r - s \equiv 6 \pmod{12}$$

From (2) and (3), we can solve for r and s :

$$r + s = 8$$

$$r - s = 6$$

Adding these equations:

$$2r = 14$$

$$r = 7$$

Subtracting the equations:

$$2s = 2$$

$$s = 1$$

So, $m \equiv 7 \pmod{12}$ and $n \equiv 1 \pmod{12}$

Now, $mn \equiv 7 * 1 \equiv 7 \pmod{12}$

To find the remainder when mn is divided by 6, we need to find:

$$7 \equiv ? \pmod{6}$$

$$7 \div 6 = 1 \text{ remainder } 1$$

Therefore, the remainder when mn is divided by 6 is 1.

Ans : 16

First, recall that $2088 = 2^3 \times 3 \times 29$

For a number to be a perfect square, all its prime factors must have even exponents.

We need to divide 2088 by a number that will leave us with even exponents for all prime factors.

Let's check each option:

a) 58:

$$58 = 2 \times 29$$

$$2088 \div 58 = (2^3 \times 3 \times 29) \div (2 \times 29) = 2^2 \times 3 = 12$$

This is a perfect square (3^2).

b) 20:

$$20 = 2^2 \times 5$$

$$2088 \div 20 = (2^3 \times 3 \times 29) \div (2^2 \times 5) = 2 \times 3 \times 29 \div 5 = 104.4$$

This is not an integer, so it can't be a perfect square.

c) 28:

$$28 = 2^2 \times 7$$

$$2088 \div 28 = (2^3 \times 3 \times 29) \div (2^2 \times 7) = 2 \times 3 \times 29 \div 7 = 74.57...$$

This is not an integer, so it can't be a perfect square.

d) 29:

$$2088 \div 29 = (2^3 \times 3 \times 29) \div 29 = 2^3 \times 3 = 72$$

This is not a perfect square.

Among the given options, only dividing by 58 results in a perfect square.

Therefore, the correct answer is 58.

Ans 17:

Here \$ represent the – operator so $X - X = 0$

First condition $x \$ (y \$ z) = x - (y - z) = x - y + z$.

It can be written as $x \$ y + z$

$$2012 - 0 + 2012 - 1912 = 2112$$

Ans : 18

First, let's understand what A and B represent:

$$A = x^3y^2$$

$$B = xy^3$$

To find the HCF (Highest Common Factor), we need to identify the common factors with the lowest exponents.

Let's break down each term:

In A (x^3y^2):

x has an exponent of 3

y has an exponent of 2

In B (xy^3):

x has an exponent of 1

y has an exponent of 3

Now, we need to choose the lower exponent for each variable:

For x, the lower exponent is 1 (from B)

For y, the lower exponent is 2 (from A)

Therefore, the HCF will be:

$$x^1y^2 = xy^2$$

Ans : 19

First, let's find the prime factorization of 1800:

$$1800 = 2^3 \times 3^2 \times 5^2$$

Now, we need to distribute these prime factors among X, Y, and Z in all possible ways.

To count the possibilities, we can use the stars and bars method. We need to distribute:

3 twos

2 threes

2 fives

For each prime factor, we have 4 possibilities (all to X, all to Y, all to Z, or split between them).

For the twos (3 of them):

$$(3+3-1) \text{ choose } 3 = {}^5C_3 = 10 \text{ ways}$$

For the threes (2 of them):

$$(2+3-1) \text{ choose } 2 = {}^4C_2 = 6 \text{ ways}$$

For the fives (2 of them):

$$(2+3-1) \text{ choose } 2 = {}^4C_2 = 6 \text{ ways}$$

By the multiplication principle, the total number of ways is:

$$10 \times 6 \times 6 = 360$$

Ans : 20

Let's say the greatest number we're looking for is x .

According to the given information:

$$148 = ax + 4 \text{ (where } a \text{ is some integer)}$$

$$246 = bx + 6 \text{ (where } b \text{ is some integer)}$$

$$623 = cx + 11 \text{ (where } c \text{ is some integer)}$$

We can rewrite these equations as:

$$148 - 4 = ax$$

$$246 - 6 = bx$$

$$623 - 11 = cx$$

This means that x must divide each of these numbers:

144, 240, and 612

We can express this mathematically as:

$$x \mid (148 - 4)$$

$$x \mid (246 - 6)$$

$$x \mid (623 - 11)$$

Now, we need to find the greatest common divisor (GCD) of these numbers:

$$\text{GCD}(144, 240, 612)$$

Let's factor these numbers:

$$144 = 2^4 \cdot 3^2$$

$$240 = 2^4 \cdot 3 \cdot 5$$

$$612 = 2^2 \cdot 3^2 \cdot 17$$

The GCD will include the common factors with the lowest exponents:

$$\text{GCD}(144, 240, 612) = 2^2 \cdot 3 = 12$$

Therefore, the greatest number that will divide 148, 246, and 623 leaving remainders 4, 6, and 11 respectively is 12.